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Abstract

This coursework has been motivated by the fact that in the digital age, data confidentiality, integrity, and confidential communication have become very important. As cyber threats advance more and more, organizations and individuals need strong cryptographic solutions ensuring that sensitive information is not accessed, tampered with, and misused by unauthorized individuals. The proposed project will be on research, development and testing of a new cryptographic system to overcome these serious security issues.

The main issue to be considered in this coursework is the necessity to have an encryption algorithm, which will enhance the current cryptographic methods in terms of stronger application of logical and mathematical operations and provide a high level of protection against the possible attacks, being efficient in terms of computation.

The coursework was methodical and entailed in-depth analysis of cryptography and learning of symmetric and asymmetric encryption algorithms, choice of a foundation cryptosystem, as well as the design and implementation of a new encryption and decryption algorithm. To ensure the algorithm was correct, efficient and secure, the algorithm was thoroughly tested with several test cases.

The project results show that the design of an efficient cryptographic system, which can ensure securing data communications and deal with the vulnerabilities of the traditional algorithms, has been achieved successfully. The project also improves the information security sphere as it offers an effective solution to secure data transfer, key management, and increased resistance to computer-based threats. Additionally, the results demonstrate how the developed algorithm could be applied to secure messaging, online banking, and storage of sensitive organizational data, as well as support its applicability in the contemporary digital security practices.

Table of Contents

1. Introduction to Cryptographic Systems	1
1.1 Aim of the Coursework	1
1.2 Coursework Objectives	2
1.3 Concept of Security	3
1.4 CIA Triad and its roles of the information security	3
1.5 Cryptography	5
1.6 History of Cryptography.....	5
1.7 Important Cryptographic Terminologies.....	6
1.8 Symmetric Encryption Systems.....	7
1.9 Asymmetric Encryption Systems.....	8
2. Background Playfair Cipher	9
2.1 Historical Background of the Playfair Cipher	9
2.2 Working on the Principle of the Playfair Cipher.	12
2.3 Algorithm of Playfair Cipher.....	12
2.4 Key Generation	13
2.5 Encryption and Decryption Example Using Playfair Cipher	13
2.6 Advantages Playfair Cipher.....	15
2.7 Disadvantage of Playfair Cipher.....	15
2.8 Where Playfair Cipher is used	16
3. Development of a L Pattern of Playfair Cipher	16
3.1 Research on Modifications to the Playfair Cipher	18
3.2 New Mathematical and Logical Operations	19
3.3 Naming the New Algorithm of L-pattern	20
3.4 Rationale for Modifications	20
3.5 New Methodology	21
3.6 The encryption process Algorithm of the L pattern	21
3.7 Decryption Algorithm process of the L pattern.....	22
3.8 Flowchart of L-pattern Algorithm	23
3.8.1 Encryption Flowchart.....	24
3.8.2 Decryption of Flowchart.....	25
4. Testing	26
4.1 Test 1.....	26
4.2 Test 2.....	29
4.3 Test 3.....	31
4.4 Test 4.....	33
4.5 Test 5.....	33
5. Evaluation.....	35

5.2 L Pattern Encryption Algorithm Weaknesses.....	36
9.3 Uses of L Pattern Encryption Algorithm.	37
6. Conclusion	37
7. References.....	38
8. Appendix	38
Appendix A	38
Appendix B	39
Appendix C	39

List of Figures

Figure 1: General Model of Cryptographic System	1
Figure 2 :cryptography security system	3
Figure 3: CIA	4
Figure 4: Cryptography	5
Figure 5: Historical Figures and Their Contributions	6
Figure 6: Symmetric encryption systems	7
Figure 7: asymmetric encryption.....	8
Figure 8: Playfair Cipher	9
Figure 9: Playfair Cipher Securing Evidence Against Threats	10
Figure 10: L Pattern Playfair Cipher.....	17
Figure 11: Modifications to the Playfair Cipher	18
Figure 12 :Mathematical and logical operations.....	19
Figure 13 :Encryption Flowchart	24
Figure 14 : Decryption of Flowchart	25

List of Table

Table 1 : Indexing of Capital Alphabets	26
Table 2 : Indexing of Small Alphabets	26
Table 3 : Indexing of Numbers.....	26
Table 4 : Indexing of Special Characters	26
Table 5 : 8*9 matrix of test 1.....	27
Table 6 : Cipher text of test 1.....	27
Table 7: : Index number of text 1	28
Table 8 : Encryption Steps of Cipher Text for Test 1	28
Table 9 : Final Decryption of Test 1	29
Table 10 : 8×9 Matrix of Keyword in L Pattern for Test 2.....	29
Table 11 : Cipher Text Table for Test 2	30
Table 12 : Index Number Encryption (i Formula) for Test 2	30
Table 13 : Decryption Using Index Formula for Test 2	30
Table 14 : Final Decryption of Test 2.....	30
Table 15 : Table 14: Matrix of Keyword in L Pattern for Test 3.....	31
Table 16 : Plain Text Pairs for Test 3	31
Table 17 : Cipher Text Table for Test 3	31
Table 18 : Index Number Encryption (i Formula) for Test 3	32
Table 19 : Decryption Using Index Formula for Test 3	32
Table 20 : Final Decryption of Test 3 Using L-Pattern Playfair Cipher.....	33
Table 21 : 8×9 Matrix of Keyword in L Pattern for Test 4.....	39
Table 22 : Cipher Text Table for Test 4	40
Table 23 : Index Number Encryption (i Formula) for Test 4	40
Table 24 : Decryption Using Index Formula for Test 4	41
Table 25 : Final Decryption of Test 4.....	41
Table 26 : 8×9 Matrix of Keyword in L Pattern for Test 5.....	33
Table 27 : Plain Text Pairs for Test 5	33
Table 28 : Cipher Text Table for Test 5	34
Table 29 : Index Number Encryption (i Formula) for Test 5	34
Table 30 : Decryption Using Index Formula for Test 5	35
Table 31 : Final Decryption of Test 5.....	35

1. Introduction to Cryptographic Systems

Contemporary information security is closely connected with the cryptographic systems that secure the data against any alteration, misuse, or exposure at any point of transmission and storage. In the digital world today, in which information is always transmitted via the Internet means of communication channels, cloud-based services, mobile applications, and online stores, people are exposed to the threats of identity theft, data breach, and unauthorized surveillance. Cryptography mitigates these threats since plain text is changed to encrypted ciphertext, such that in case of interception, the content cannot be read and decrypted only with the right key. It has a crucial role in the preservation of privacy, data integrity verification and authentication. The most frequently used core principles such as CIA triad- Confidentiality, Integrity and Availability act as the foundation of secure systems and both asymmetric encryption and symmetric encryption offer effective solutions that can be used to protect information. Since ancient secret codes to modern complex algorithms, cryptography has become an essential part of cybersecurity, and **Example:** WhatsApp such practical cases like end-to-end encryption in WhatsApp have guaranteed that only the target audience can understand the messages (Richards, 2024).

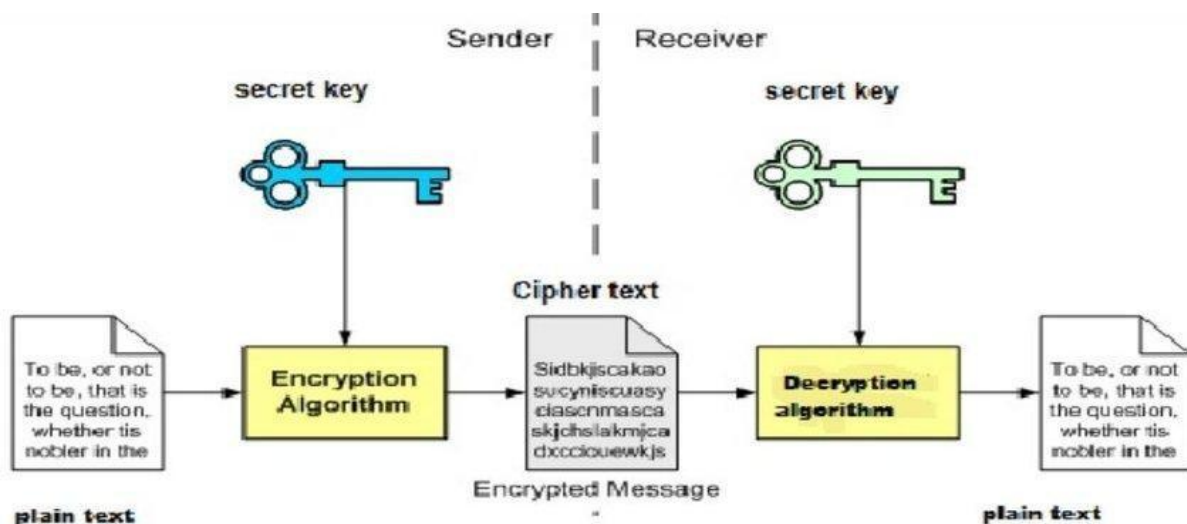


Figure 1: General Model of Cryptographic System

1.1 Aim of the Coursework

The general goal of this course work is to investigate, design, develop and test a new cryptographic system, based on the available cryptographic concepts and applied new of mathematical and logical functions, in order to provide a feasible solution to the problem of maintaining data and information integrity and privacy of the computing

1.2 Coursework Objectives

- To conduct an in depth study of cryptography, including its history, symmetric cryptographic techniques, and asymmetric cryptographic techniques, in order to build a strong theoretical foundation
- To select a cryptographic algorithm that has already been created and developed and analyse it.
- To design and generate a novel cryptographic algorithm, the base algorithm selected requires the addition of novel mathematical and logical operations.
- Encryption and Decryption facilities to be installed to the recently developed cryptographic system to justify reasonable workability.
- To test the developed cryptography algorithm and see that it is correct, reliable, and works correctly a certain number of test cases should be verified.
- To evaluate the pros and cons of the proposed cryptographic system, one should figure out the benefits and flaws of the system, and how it may be used.
- To prepare a detailed research, development, testing and evaluation project which is captured in a professional technical report that is academic and professional nature.

1.3 Concept of Security

Security in computing is a concept of ensuring the safeguarding of information systems and networks against an unauthorised access, alteration or destruction. It also makes sure that sensitive information is confidential, correct and accessible to authorized personnel.



Figure 2 :cryptography security system

Security plays the an important role of ensuring continuity in the operations of the an organization, privacy and adherence to laws and regulations in the context of contemporary an organizations. Good security controls deter cyberattacks, information breaches, and insider an attacks that can threaten the integrity of information systems.**Examples:** A bank deploys multi-factor of authentication, firewalls, and intrusion detection systems so that the customer data, transaction records and internal systems are not accessed by unauthorized individuals.

1.4 CIA Triad and its roles of the information security

One of the most basic models of information security interpretation and application is CIA Triad. It is composed of three principles:

- **Confidentiality:** Confidentiality helps to maintain information to those individuals with an authorized access. Methods like encryption, access control lists and secure authentication systems ensure confidentiality. **Examples:** Secrecy of medical records by encrypting them such that they are only seen by the doctor or patients except when the doctor needs to access.
- **Integrity:** Integrity ensures that data does not get distorted or changed when storing or transmitting it. Checksums, digital signatures and hashing are some of the mechanisms that are used to ensure that information is not distorted. **Example:** To calculate whether a file has been altered, it is possible to use digital signatures or hash functions
- **Availability:** Availability is a good guarantee that data and resources can be accessed by an authorized user when required. Maintenance is done via redundancy, backups and disaster recovery plans to ensure availability. **Example:** Cloud services: backup servers and the redundancy to ensure access in an instance of server failure.



Figure 3: CIA

The CIA Triad is important in providing information security that is of quality in that failure in one of the three concepts can jeopardize the overall security position of an organization (Contributors, 2025).

1.5 Cryptography

Cryptography refers to science and an art of ensuring that information is secured by changing it into a non-readable form by unauthorized users. It is a critical element of information security that has confidentiality, integrity, authentication and non-repudiation.



Figure 4: Cryptography

Example : Secure messaging systems, online banking, digital signatures, and Virtual Private Networks (VPNs) are some of the applications of cryptography (Hangrae, 2024).

1.6 History of Cryptography

Cryptography has been developed throughout centuries as the means of ensuring the safety of communication and making sure that sensitive information is not accessed by unauthorized parties. Early cryptographic methods in ancient civilizations like Mesopotamia and the Egypt were based on the simple substitution ciphers in which messages were hidden by replacing letters or symbols with others. Under the Roman rule, Julius Caesar came up with the caesar cipher whereby the letters in the alphabet were moved by the predetermined number to code military and political messages.



Figure 5: Historical Figures and Their Contributions

Later in the Middle Ages, more advanced methods such as the polyalphabetic Vigenere Cipher were devised in order to resist attacks of frequency analysis. The strategic significance of secure communication was proved during the World War Era when the armies deployed an encrypted method of messages on the large scale using mechanical cipher machines as, **e.g.**, the Enigma. Digital **cryptography (RSA, AES, and ECC)** is used in the contemporary world to provide safe network data transmission e-commerce, and safe digital information in the modern world. As an example, online banking of systems encrypt the data with the help of AES, making the information of the customers confidential and the safe.

1.7 Important Cryptographic Terminologies.

To learn about cryptography, one should be aware of some terminology:

- **Plaintext**- The initial readable information which should be secured.
- **Ciphertext**- The encrypted form of data, which is unreadable.
- **Encryption** -The transformation of plaintext into ciphertext.

- **Decryption**- The reciprocal action, whereby cipher text is transformed into plain text.
- **Key** -A coded number that is used to encrypt and decrypt.
- **Cryptosystem** - The general structure of algorithms and techniques used to process the encryption and decryption.
- **Symmetric Key**- a single key is applied in encryption and decryption.
- **Asymmetric Key** -Two keys (public and private) are used simultaneously to encrypt and decrypt.
- **Hash Function** - This is a mathematical function that generates a fixed length hash (to ensure integrity of data).

1.8 Symmetric Encryption Systems

Symmetric encryption refers to an encryption technique in which both the sender and receiver of the message employ the same secret key to encrypt and decrypt the data respectively. This implies that the sender and receiver should share a common key. It is quick and effective, and thus it can be used on the encrypting of a huge mass of data. The most widespread symmetric algorithms are AES that is actively used nowadays and the outdated DES that is no longer active because of its vulnerability to attacks. Speed is the primary benefit of symmetric encryption, whereas the primary drawback is the key distribution as the data can be revealed in case the key is stolen. As an illustration, cloud storage systems apply AES to encrypt files such that only authorized users can decode it (GeeksforGeeks, 2025).

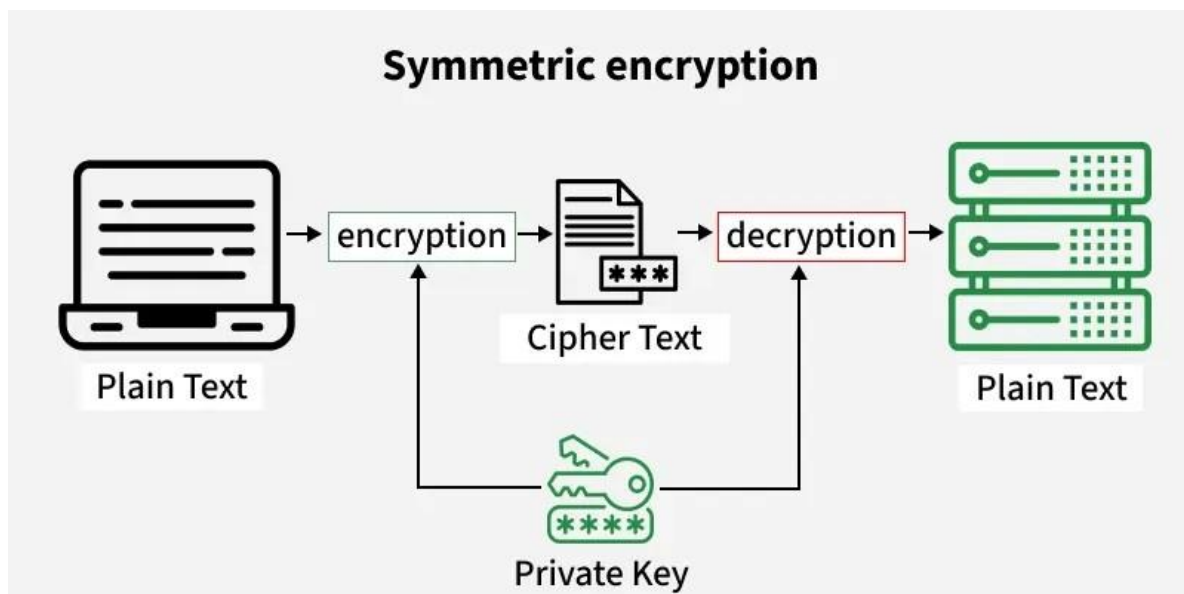


Figure 6: Symmetric encryption systems

1.9 Asymmetric Encryption Systems

Public-key cryptography is also known as asymmetric encryption, which employs two keys, one of which is the public key used in encryption and the other is the private key used in decryption. This is the solution to the key-sharing issue of symmetric encryption since it is safe to share the public key. It is also compatible with authentication using digital signature, message integrity and verification of the sender. The widely used ones are RSA and Elliptic Curve Cryptography (ECC) which offers high security with a smaller key size. Asymmetric encryption is slower than symmetric encryption and is normally employed in securing the transmission of encryption keys, e.g. in secure email systems. Despite being safer than key exchange, asymmetric encryption is still slower.

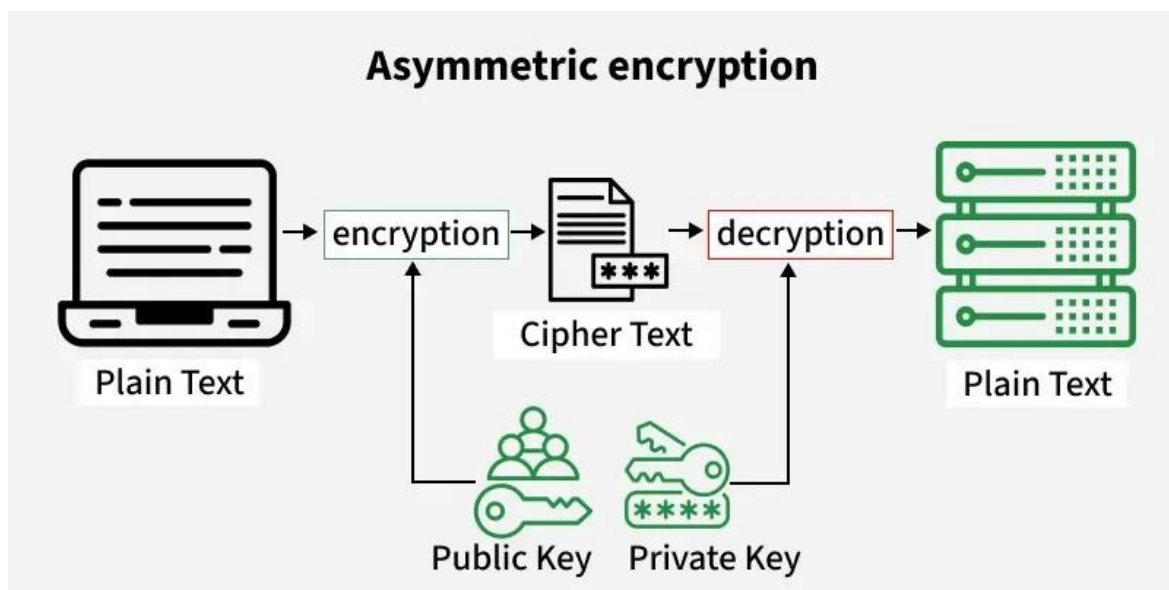


Figure 7: asymmetric encryption

2. Background Playfair Cipher

Playfair Cipher is a primitive method of encrypting a text message, which uses pairs of letters as opposed to individual letters. This renders it more impregnable than mere substitution codes and more difficult to crack through frequency analysis. It works with a 5 by 5 matrix that is formed out of a keyword and the letters I and J are merged to form the alphabet. Playfair Cipher was significant in the earliest history of cryptography (Nagaraj, 2023).

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Figure 8: Playfair Cipher

2.1 Historical Background of the Playfair Cipher

The Playfair Cipher is a notion of classical cryptography and is one of the oldest and most commonly known manual methods of symmetric types of encryption. It takes an important position since it was the first real digraph substitution cipher; that is,

pl	p	r	e	b	l
	e	m	a	c	d
	f	g	h	i	k
	n	q	s	t	u
	v	w	x	y	z

ay	p	r	e	b	l
	e	m	a	c	d
	f	g	h	i	k
	n	q	s	t	u
	v	w	x	y	z

fa	p	r	e	b	l
	e	m	a	c	d
	f	g	h	i	k
	n	q	s	t	u
	v	w	x	y	z

ir	p	r	e	b	l
	e	m	a	c	d
	f	g	h	i	k
	n	q	s	t	u
	v	w	x	y	z

Figure 9: Playfair Cipher Securing Evidence Against Threats

Origin and Development

In 1854, Playfair Cipher was created by a British scientist and inventor, Charles Wheatstone, who was known largely due to his efforts in electrical engineering and telegraphy. Though it was Wheatstone who designed the cipher, it was given the name of Lord Lyon Playfair who was a British politician and a scientist who made it highly popular.

Lord Playfair realized the need to have tight communication particularly in military and diplomatic matters. Due to his work, the cipher came to be commonly referred to as Playfair Cipher, despite Wheatstone being the founder.

Purpose and Motivation

In the middle of the 19th century, the majority of encryption schemes involved simple replacement with a single letter, and the Caesar Cipher was one of them. Frequency analysis was simple to crack these techniques since some letters (such as E, T, A) are more common in English.

To address this weakness the the PlayfairCipher was created that encrypted two letters at once. In this way, it conceals the individual frequencies of letters and complicates cryptanalysis particularly when performed manually without the use of computers.

Military Adoption and Usage

The Playfair Cipher was later adopted although initially the British government deemed it too complex to use in the field, it was adopted because of its security versatility.

Playfair Cipher was mostly applied in:

- British Army
- British Navy
- World War I
- World War II
- Boer War

It was specifically popular due to the reason that:

It could be performed by hand

No machines were required

It was quick enough to communicate on the battlefield.

It was stronger compared to plain substitution ciphers.

Example of Historical Use

The Playfair Cipher was employed during the World War I to encode battalion commanders messages. Although it was later cracked by the experts of the art of cryptotherapy, it gave a reasonable level of protection to postpone the awareness of the enemies, which, in many cases, gave adequate time during wartime.

Cryptographic Significance

Historical importance is the fact that the Playfair Cipher is historically significant because:

- It put forward the idea of polygraphic substitution.
- It led to the subsequent and more complicated encryption methods.
- It proved that encryption in pairs of letters contributes greatly to security enhancement.
- It was a transition between primitive ciphers and the current cryptography.

Decadence and Contemporary View.

Playfair Cipher cannot be regarded as safe in the contemporary communication because of the development of computers and methods of cryptanalysis. The method of breaking it can today be carried out with the help of frequency analysis computer based (Schneider, 2025).

2.2 Working on the Principle of the Playfair Cipher.

The Playfair Cipher works on a few basic principles which render it a classical symmetric encryption method. It is a symmetric key crypt, that is, the same key is employed in encryption and decryption, and so, only the parties holding the common key can get the message. In contrast to the basic substitution ciphers, the Playfair Cipher uses digraph substitution, in which the plaintext is broken into pairs of letters, known as digraphs and the pair is encrypted as a unit. A randomized key of 5x5 is created with the help of a selected keyword, consisting of all letters of the alphabet (typically mixing I and J together). The encryption procedure is governed by certain rules: When there are two letters at a digraph in the same row, both letters are substituted by the letter at its right and when there are two letters in the same row in the column, the letters are substituted by the letters at the same row in the column of the other letter. To deal with any special cases, the cipher employs padding rules: repeating letters in a pair are divided by an X, and when the number of letters in the plaintext is an odd number, then an X is tacked at the end to finish the last digraph. All these principles combined to bring the Playfair Cipher more secure than plain monoalphabetic ciphers and at the same time more manageable to encrypt manually.

2.3 Algorithm of Playfair Cipher

Playfair Cipher algorithm is a systematic sequence of steps in encrypting and decrypting messages safely. To begin with, a key word is selected as a secret one, and it is used to create the encryption matrix. A 5x5 matrix of letters is created with the letters of the keyword and the rest of the letters of the alphabet, the letters I and J usually are combined to use all the letters. The plaintext is also preprocessed including converting it to uppercase, eliminating spaces, replacing any incidence of J with I and adding a filler letter (typically X) between the repeated letters in a pair to prevent any duplication. The real encryption code is done on pairs of letters in the form of digraphs using the Playfair rules, the letter in the same row is substituted by the letter to the right of the message, the letter in the same column is substituted by the letter below,

and the corner letters of the message are substituted by the letters in the same row but the column of the letter in the other message. In the case of decryption, these rules are reversed and by following the same key matrix 5x5, the original plaintext can be reassembled.

2.4 Key Generation

Key: MONARCHY

- To obtain the key matrix, the following steps are to be followed:
- Write the keyword in lower case without repetition.
- Write in any letters of the alphabet (do not include J).
- Combine I and J

Generated 5x5 Matrix

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Both encryption and decryption are done using this matrix.

2.5 Encryption and Decryption Example Using Playfair Cipher

Playfair Cipher Basics

- Uses a 5x5 matrix of letters (I and J are usually combined).
- Encryption is done in digraphs (pairs of letters).
- Rules

1. Same row – each letter replaced by one to its right (rolling over if necessary).

2. Same column → replace each letter by letter by one position down.

3. Rectangle rule -> replace each letter by replace each letter with the letter in the same row, but in the column with the other letter.

Step1: Construct the Key matrix

For instance, assume that "MONARCHY"

Matrix (I/J)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Step 2: Prepare the Plaintext

Plaintext: "INSTRUMENTS"

- Break into digraphs

INSTRUMENTS

- If the pair consists of repetitive letters, add X (not required in this case).
- If it's odd length, append X to the end (which is already even).

Step 3: Encryption

Apply Playfair rules:

1. **IN** → I and N combine to make a rectangle → **G** and J (J = I, so → **G** and N's column → **G** and N's row → result: GM)

IN: I (row 3, col 4), N (row 1, col 3) → rectangle → G (row 3, col 3) and A (row 1, col 4) → GA

- **ST**: S (row 4, col 4), T (row 4, col 5) → same row → TX → **TU**
-

RU: R (row 1, col 5), U (row 5, col 1) → rectangle → M (row 1, col 1) and Z (row 5, col 5) → MZ

- **ME**: M (r 1, c 1), E (r 3, c 1) → Same Column: C (r 2, c 1), L (r 4, c 1) → CL

- **NT:** N (r1, c3), T (r4, c5) → rectangle → R (r1, c5) & Q (r4, c3) → RQ
- **SX:** S (Row 4, Col 4), X (Row 5, Col 4) → same column → I (Row 3, Col 4), S (Row 4, Col 4) → IS

Ciphertext: **GA TU MZ CL RQ IS**

Step 4: Decryption

The Reverse the Rules:

- **GA** → rectangle back → IN
- **TU** → same row left shift → ST • **MZ** → rectangle back → RU •
- **CL** → same column up shift → ME • **RQ** → rectangle back → NT • **IS** → IS same column up shift → SX → original padding removed → S

Plaintext = INSTRUMENTS

2.6 Advantages Playfair Cipher

- Ciphers two letters (digraphs) simultaneously, thus being more secure compared to monoalphabetic ciphers, like Caesar Cipher
- Eliminates the power of single-letter frequency analysis which was one of the greatest weaknesses of the previous ciphers.
- Simple enough to be understood and applied even without high mathematical skills.
- Appropriate to be encrypted and decrypted manually, particularly where it is not digital.
- Simply needs a simple keyword to generate keys.
- Historical as it was one of the earliest practical polygraphic ciphers.
- Academically relevant since it is often used in exams, coursework and in questions during the viva.
- Develops a solid base on learning of symmetric key cryptography.

2.7 Disadvantage of Playfair Cipher

- Nonetheless, can still be subject to analysis of digit frequency provided that there is adequate ciphertext.

- Has a small key space, which is why it is not as strong as contemporary crypto algorithms.
- Weak to present cryptanalytic attacks and processing.
- Requires careful plaintext preprocessing (handling repeated letters, padding with X, replacing J with I).
- The predictability of rules in encryption is given that one knows the key matrix.
- Unable to manipulate numbers, symbols and other special characters.
- Unsuitable in the digital age and the real world of security.

2.8 Where Playfair Cipher is used

- Education and academic courses on the teaching of the classical encryption concepts.
- Computer security, information security and networking coursework and exams.
- To learn about historical development on cryptographic methods.
- This research is in military communication and historical cryptography.
- To design modified or enhanced cryptographic systems as a foundation algorithm in research projects.
- Applied to real-life examples of encryption and cipher designing rules based on symmetric key encryption.

3. Development of a L Pattern of Playfair Cipher

The approach enhances security due to the enhanced complexity despite being manual. With the comparison to the usual Playfair Cipher, the L-Pattern method cryptanalysis becomes more challenging due to the less predictable substitutions. The reversal of the same L-shaped rules is done to do the decryption. Altogether, L-Pattern Playfair Cipher strengthens the key cipher and thus, remains plain and effective.

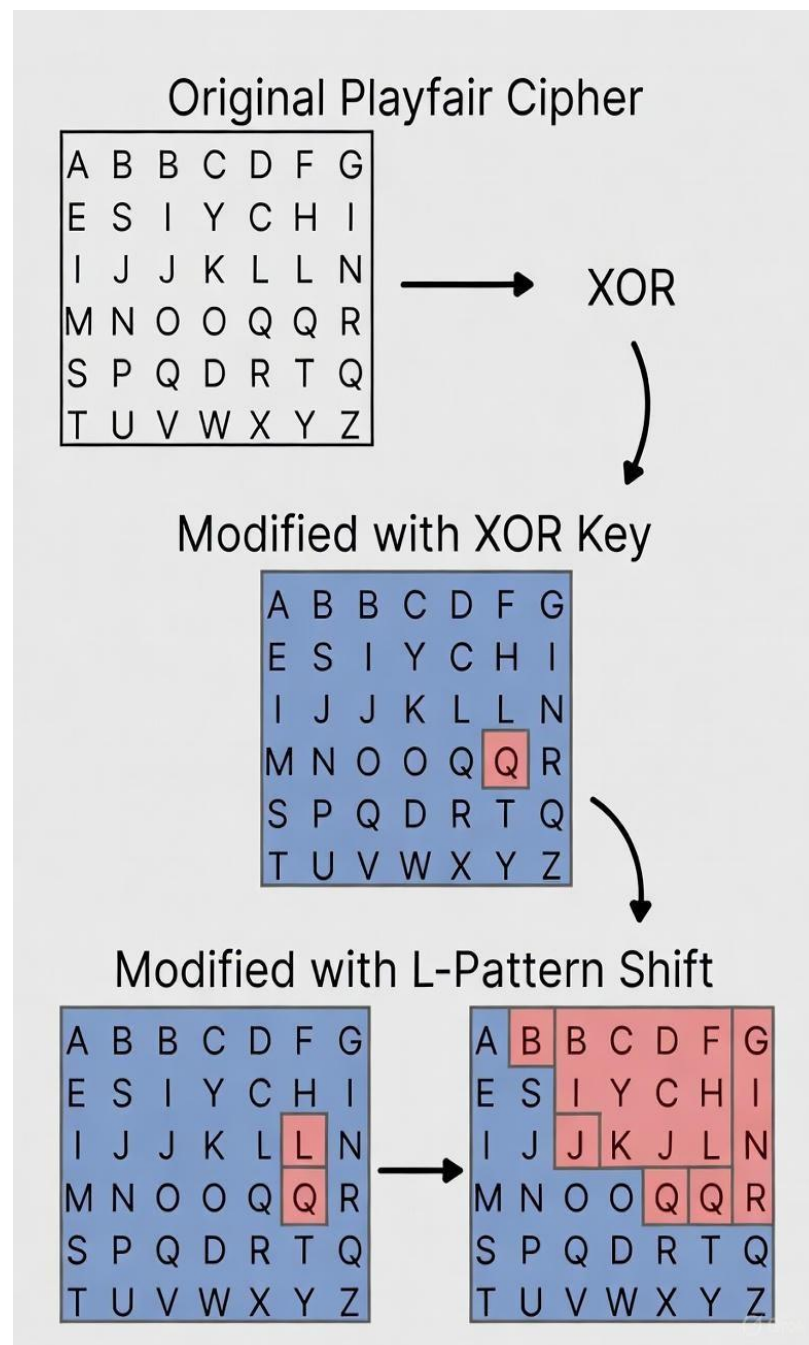


Figure 10: L Pattern Playfair Cipher

The approach enhances security due to the enhanced complexity despite being manual. With the comparison to the usual Playfair Cipher, the L-Pattern method of cryptanalysis becomes more challenging due to the less predictable substitutions. The reversal of the same L-shaped rules is done to do the decryption. Altogether, L-Pattern Playfair Cipher strengthens the key cipher and thus, remains plain and effective.

3.1 Research on Modifications to the Playfair Cipher

The Playfair Cipher has been studied extensively to detect and make required changes to make the cipher more resilient to frequency analysis, known-plaintext attacks as well as brute force techniques. A common method is to make the key bigger (10x10 or bigger) to add more characters (such as digits and symbols) and thus more complex to the substitution process making it less predictable. The other change involves creating several iteration steps and different keys, which involves repeating the encryption step to enhance the ciphertext with regard to the differential attack. Pseudorandom number generators including a modified version of the Blum Blum Shub algorithm have been considered to generate dynamic keystreams resulting in forward secrecy and interfere with fixed patterns in the cipher. The permutations in the matrix or the change in the processing of digraphs like adding new variable change or bit operations, are additional to obscure the digraph frequencies and promote diffusion. All these adjustments are meant to prevent the weaknesses of the Playfair Cipher and make it useful in terms of speed and manual or low-resource environments (Shakil, 2014).

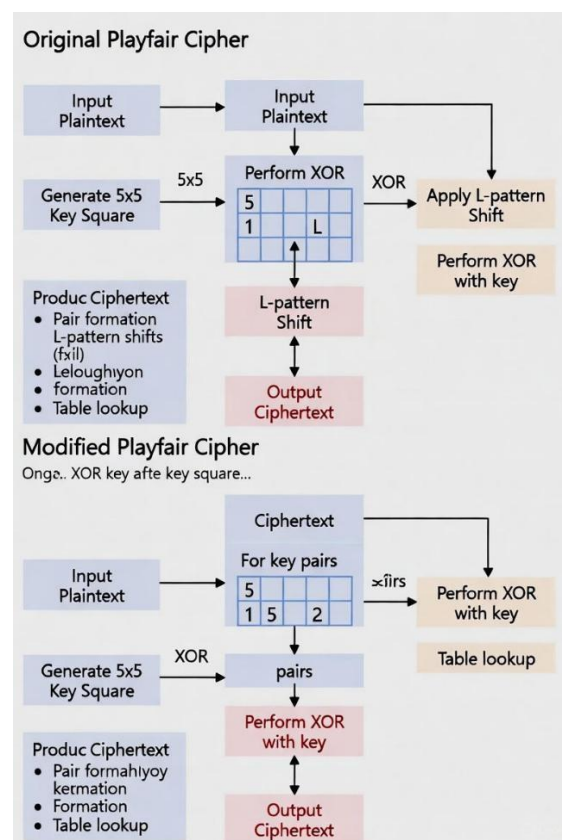


Figure 11: Modifications to the Playfair Cipher

3.2 New Mathematical and Logical Operations

The L-pattern algorithm adds a number of mathematical and logical operations to enhance security all of which have a solid foundation in the field of cryptography. L-shaped shift is a geometrical operation which corresponds to non-linear steps in variant ciphers, such as the knight moves in chess, with the difference that it moves 2 positions along one direction (horizontal or vertical) and 1 position perpendicular to it, with modular arithmetic used to roll-off around the edges of the matrices. This step increases the confusion in that the original Playfair is rectangular in form. Bitwise logic XOR (exclusive OR) is a bit-wise logical operation which is used to combine binary representations of position values or ASCII codes with bits derived by the key, and which is found in contemporary ciphers such as AES to do diffusion, with a single bit alteration in the key influencing multiple encrypted bits. The modular arithmetic (modulo 5, being 5x5 grid) provides cyclic behavior in shifts, which is a concept of number theory that is used in most substitution ciphers to manage boundaries. Another logical operation in binary arithmetic, parity checking (checking whether the length of a keyword is even or odd) is a binary arithmetic operation that provides variability in shift directions, like conditional branching in pseudorandom generators. All these types of operations make the algorithm more and more resistant to statistical attacks, introducing complexity and randomness.

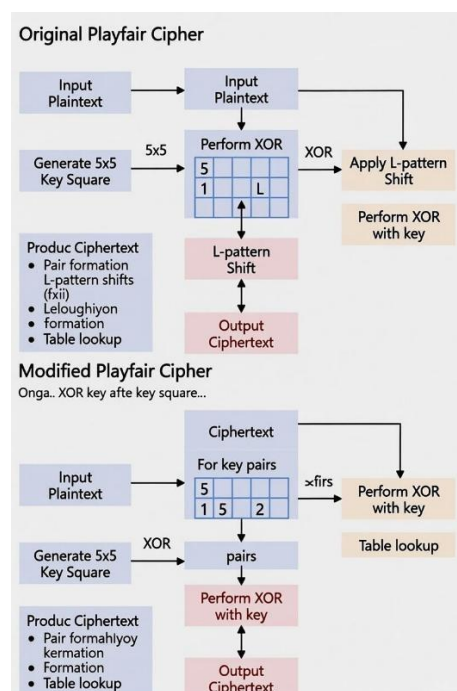


Figure 12 :Mathematical and logical operations

3.3 Naming the New Algorithm of L-pattern

The new cryptographic algorithm was named the L-pattern so as to reflect its novelty in the basic sense the use of L-shaped shifts in the replacements process. This nomenclature is based on the geometrical arrangement of the shifts, which resemble the shape of an L, i.e., 2 steps in a given direction then 1 step in a right angle, and thus is not the rectangular or linear patterns of the ancient Playfair Cipher. The name represents a conceptual leap in the design of security, with much focus on angular, non-linear traversal to create asymmetry and uncertainty, just as other variants of the cipher are named after key attributes (e.g. the Four-Square Cipher is named so because of its grid formation). Choosing L-pattern as the name of the algorithm emphasizes that the algorithm aims at improving diffusion by using patterned movements, and it would be intuitive to the cryptographers to compare the name with the essential mechanism of the algorithm and stand it in contrast with variants that keep the rectangular substitutes.

3.4 Rationale for Modifications

These changes to the Playfair Cipher were necessary to address some of the known weaknesses in this algorithm that make it unsuitable to the modern security requirements. First of all, the conventional Playfair will be vulnerable to digraph frequency analysis, where frequent pairs of letters such as "TH" or "HE" create recognizable patterns in the encrypted message and allow the attacker to use statistical distributions to decode the message if you read more then click on [Appendix A](#). These improve the cryptanalysis to be more computationally intensive, which makes it more resistant to brute-force and are statistical attacks, without being overly complicated to base structure.

3.5 New Methodology

The novel approach to the L-pattern algorithm retains all the classical components of the Playfair Cipher including the square of five digits (key) and the processing of digraphs (via digraph shifts), but introduces the new principle of the L-shaped shifts of digraphs that do not coincide in the same row or column. This change is by computing new positions by a pattern of two steps horizontally, then one step vertically (or the reverse), modulated by the evenness or oddness of the keyword length, even length uses horizontal-first, odd vertical-first, to add some variability contingent on conditioning, and some predictability if you read more then click on [Appendix B](#). This demonstrates that non- linearity and diffusion are imposed on the methodology so that the same digraphs used in different scenarios will yield different outputs.

3.6 The encryption process Algorithm of the L pattern

Step 1: Start

STEP 2: Take input plaintext and keyword

STEP 3: Initialize ciphertext with an empty string

STEP 4: Create a 5x5 key square from the keyword

- Combine I/J into a single cell

STEP 5: Preprocess the plaintext

- Case conversion
- Convert all letters to uppercase.
- Remove non-alphabetic characters
- Insert 'X' between repeated letters in a digraph
- Append 'X' if plaintext length is odd

STEP 6: Creating the XOR key by translating the characters of the keyword into ASCII and cycling them

STEP 7: Break down the prepared plaintext into digraphs.

STEP 8: In each digraph, find the positions of both letters in the key square.

STEP 9: If both are in the same row, move each letter to the right (mod 5).

STEP 10: Else: If both letters are located within the same column, move both letters down modulo 5.

STEP 11: Else apply parity-based L-shift

Horizontal-Vertical (H-V) or Vertical-Horizontal (V-H) depending on position

STEP 12: XOR operation on the row and column indexes using the XOR key (mod

STEP 13: Perform lookups in the key square using the obtained positions to get the encrypted letters

STEP 14: The encrypted digraph is appended to the ciphertext.

STEP 15: For all Digraphs, repeat steps 8-14

STEP 16: Print the final ciphertext

STEP 17: Return or store the encrypted message

STEP 18: End

3.7 Decryption Algorithm process of the L pattern

STEP 1: Start

STEP 2: Input the ciphertext and the keyword

STEP 3: Set plaintext equal to an empty string

STEP 4: Creation of the same 5×5 key square using the keyword

STEP 5: The XOR key must be generated in the same way as in the encryption process

STEP 6: The ciphertext is split into digraphs

STEP 7: Now, for each digraph, determine where each row and column intersects in the key square. The reason this works is that each

STEP 8: Invert the XOR on positions (mod 5)

STEP 9: If both letters are in the same row, move both letters to the left (mod 5)

STEP 10: letters are in same column, move each letter up mod 5

STEP 11: Else reverse L-shift based on parity

- H-V Shift: V-H Shift
- The value

STEP 12: Search these positions in the key square for the decrypted letter.

STEP 13: Add the decrypted digraph to the plaintext

STEP 14: Proceed to step 7 and repeat for all digraphs

STEP 15: Eliminate padding character 'X' (if any)

STEP 16: Printing the final plaintext:

STEP 17: Return or store the decrypted message

STEP 18: End

3.8 Flowchart of L-pattern Algorithm

The flowchart in the algorithm of the L-pattern has been split into two separate diagrams, one for decryption and one for encryption.

3.8.1 Encryption Flowchart

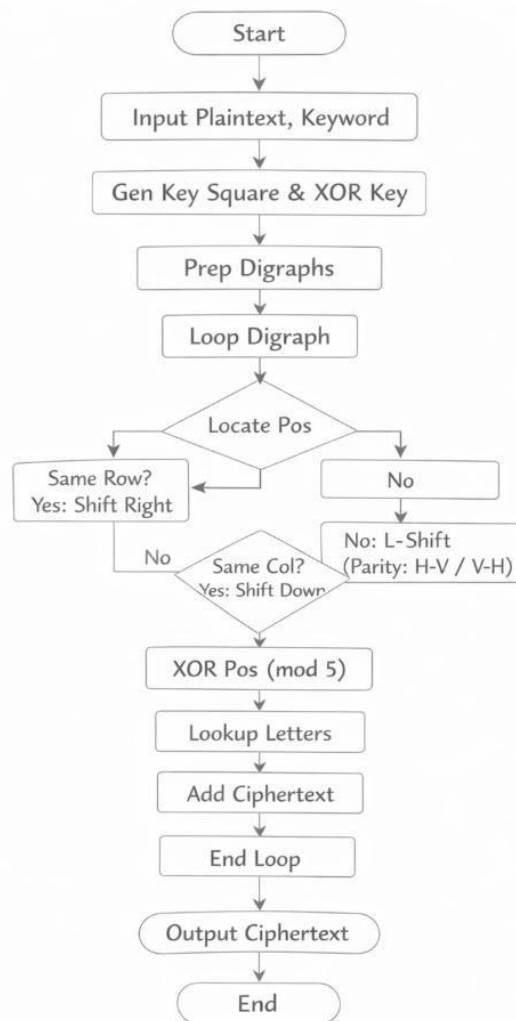


Figure 13 :Encryption Flowchart

3.8.2 Decryption of Flowchart

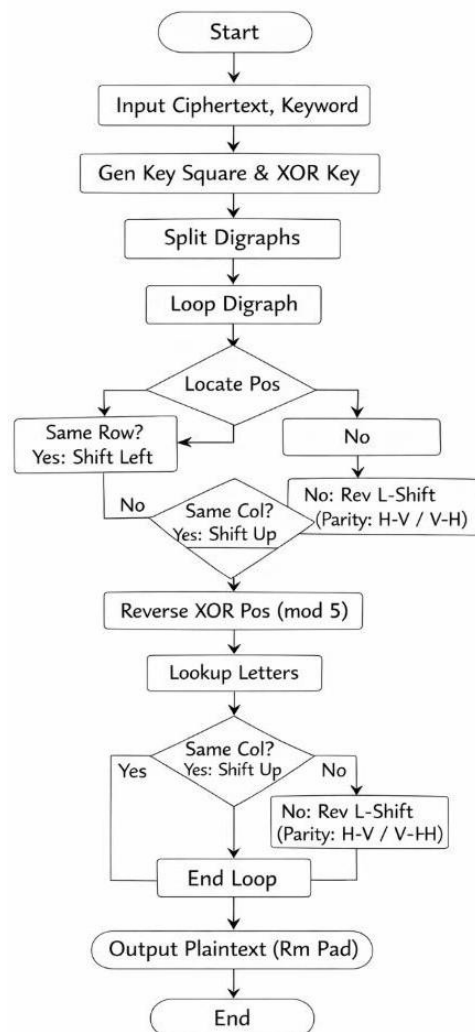


Figure 14 : Decryption of Flowchart

4. Testing

The L Pattern Playfair Cipher algorithm will be tested on five tests such as creation of index number that is made of capital letters, small letters, numbers, and special characters to make sure that it is efficient in the encryption and decryption.

Ind ex	0	1	2	3	4	5	6	7	8	9	1 0	1 1	1 2	1 3	1 4	1 5	1 6	1 7	1 8	1 9	2 0	2 1	2 2	2 3	2 4	2 5
Ch ar	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Table 1 : Indexing of Capital Alphabets

Ind ex	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Ch ar	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z

Table 2 : Indexing of Small Alphabets

Index	0	1	2	3	4	5	6	7	8	9
Number	0	1	2	3	4	5	6	7	8	9

Table 3 : Indexing of Numbers

Inde x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Char	!	@	#	\$	%	&	*	(	)	_	+	=	?	,	.	SPA CE

Table 4 : Indexing of Special Characters

4.1 Test 1

Plain Text: Hello World!

Key : Cipher@123

Solution:

Making matrix of the 8×9 using the keyword in L Pattern.

C	i	p	h	e	r	@	1	2
3	H	e	l	l	o	W	o	r
l	d	!	A	B	C	D	E	F
G	l	J	K	L	M	N	O	P
Q	S	T	U	V	X	Y	Z	a
b	c	f	g	j	k	m	n	q
s	t	u	v	w	x	y	z	0
C	i	p	h	e	r	@	1	2

Table 5 : 8*9 matrix of test 1

Step 1: Divide a Plain Text into Pairs

He ll o<space> Wo rl d!

Plain Text Pair	Encryption Rule Applied	Cipher Text
He	Different Row & Column (Rectangle Rule)	l
ll	Same Row (Shift Right)	@
o <space>	Different Row & Column (Rectangle Rule)	E
Wo	Different Row & Column (Rectangle Rule)	l
rl	Same Row (Shift Right)	3
d!	Different Row & Column (Rectangle Rule)	W
		p
		2
		r
		!

Table 6 : Cipher text of test 1

Step 2: Apply the Encryption of Rules to Each Pair

Same Row: Shift each character to the **right** in the same row (wrap around if at the end).

Same Column: Shift each character **down** (wrap around to the top if at bottom).

Different Row & Column: Form a **rectangle**, replace each character with the letter in its row but column of the other character.

Final Cipher Text: l@El3Wp2r!

Position	1	2	3	4	5	6	7	8	9	10
Cipher Character	l	@	E	l	3	W	p	2	r	!

Table 7: :Index number of text 1

Table 8: Index Number Encryption (i Formula)

Rules:

- **Odd index (i):** $(i+3)/2$ $(i+3)/2$ $(i+3)/2$
- **Even index (i):** $i/2$ $i/2$ $i/2$

Applying formula on Cipher Text (l@El3Wp2r!):

Character	Index (i)	Formula	Resulting Char
l	11	$(11+3)/2 = 7$	H
@	6	$6/2 = 3$	#
E	4	$4/2 = 2$	C
l	11	$(11+3)/2 = 7$	H
3	3	$(3+3)/2 = 3$	3
W	22	$22/2 = 11$	L
p	15	$(15+3)/2 = 9$	J
2	2	$2/2 = 1$	1
r	17	$(17+3)/2 = 10$	K
!	0	$0/2 = 0$	!

Table 8 : Encryption Steps of Cipher Text for Test 1

Encrypted Text after Indexing: H#CH3LJ1K!

Decryption Process: Reverse Index Formula

Rules:

Odd index: $i*2-3$ $i*2-3$ $i*2-3$

Even index: $i*2$ $i*2$ $i*2$

Reversing H#CH3LJ1K!:

Character	Index (i)	Formula	Original Char
H	7	$7*2-3 = 11$	l
#	3	$3*2 = 6$	@
C	2	$2*2 = 4$	E
H	7	$7*2-3 = 11$	l

3	3	$3*2-3 = 3$	3
L	11	$11*2-3 = 19$	W
J	9	$9*2-3 = 15$	p
1	1	$1*2 = 2$	2
K	10	$10*2-3 = 17$	r
!	0	$0*2 = 0$	!

Table 9 : Final Decryption of Test 1

Recovered Cipher Text: l@El3Wp2r!

Table 9: Final Decryption of Test 1

Final Plain Text: Hello World!

4.2 Test 2

Plain Text: Code

Key: Cyber123!

Solution:

Making an **8×9 matrix** of keyword in **L Pattern**.

Table 10: 8×9 Matrix of Keyword in L Pattern for Test 2

G	y	C	1	2	3	&	?	!
c	b	D	E	F	0	.	SPACE	@
r	t	H	I	J	K	L	M	N
e	p	O	P	Q	R	S	T	U
d	f	V	W	X	Y	Z	a	b
g	h	i	j	k	l	m	n	o
s	u	v	w	x	y	z	A	B
!	#	\$	%	^	*	(	)	-

Table 10 : 8×9 Matrix of Keyword in L Pattern for Test 2

Step 1: Split Plain Text into Pairs

Co de → Co, de

Step 2: Encrypt the Pairs Using L-Pattern Rules

- **Co** → H@ (Rectangle Rule, different row & column)
- **de** → l2 (Rectangle Rule, different row & column)

Cipher Text: H@l2

Cipher Text Table for Test 2

Plain Text Pair	Cipher Text
Co	H@
de	l2

Table 11 : Cipher Text Table for Test 2

Final Cipher Text: H@l2

Table 12: Index Number Encryption (i Formula)

Rules:

- **Odd index (i):** $(i+3)/2$ (i + 3)/2 (i+3)/2
- **Even index (i):** $i/2$ / $2i/2$

Cipher Char	H	@	l	2
Original Index (i)	7	1	11	2
Formula Applied	$(7+3)/2$	$(1+3)/2$	$(11+3)/2$	$2/2$
Encrypted Index	5	2	7	1
Encrypted Char	E	#	H	1

Table 12 : Index Number Encryption (i Formula) for Test 2

Encrypted Text after Indexing: E#H1

Decryption Process (Reverse i Formula)

Encrypted Char	E	#	H	1
Encrypted Index	5	2	7	1
Reverse Formula	$(5*2-3)$	$(2*2)$	$(7*2-3)$	$(1*2)$
Original Index	7	4	11	2
Cipher Char after L-Pattern	H	@	l	2

Table 13 : Decryption Using Index Formula for Test 2

Decrypted Text after Indexing: H@l2

Final Step: Reverse L-Pattern pair rules → **Plain Text:** Code

Cipher Text Pair	Decrypted Pair
H@	Co
l2	de

Table 14 : Final Decryption of Test 2

4.3 Test 3

Plain Text: Secure123

Key: KeyPlay2026!

Solution: Making an **8×9 matrix** of keyword in **L Pattern**.

Matrix of Keyword in L Pattern for Test 3

K	e	y	P	I	a	y	2	0
6	!	S	E	C	U	R	1	3
4	5	7	8	9	&	*	@	#
Q	W	O	T	I	N	M	L	H
F	G	A	B	D	X	Y	Z	c
v	w	x	y	z	o	p	q	r
s	t	u	m	n	b	f	g	h
SPACE	.	,	?	(	)	+	=	%

Table 15 : Table 14: Matrix of Keyword in L Pattern for Test 3

Plain Text Pairs for Test 3

Plain Text	Split into Pairs
Secure123	Se, cu, re, 12, 3x

Table 16 : Plain Text Pairs for Test 3

Cipher Text Table for Test 3

Plain Text Pair	Cipher Text
Se	Q?
cu	3H
re	l2
12	@6
3x	!Z

Table 17 : Cipher Text Table for Test 3

Final Cipher Text: Q?3Hl2@6!Z

Index Number Encryption (i Formula) for Test 3

Rules:

- **Odd index (i):** $(i+3)/2(i+3)/2(i+3)/2$

- Even index (i): $i/2i / 2i/2$

Cipher Char	Q	?	3	H	I	2	@	6	!	Z
Original Index (i)	16	5	3	7	11	2	1	6	0	25
Formula Applied	$(16/2)$	$(5+3)/2$	$(3+3)/2$	$(7+3)/2$	$(11+3)/2$	$2/2$	$(1+3)/2$	$6/2$	$(0/2)$	$(25+3)/2$
Encrypted Index	8	4	3	5	7	1	2	3	0	14
Encrypted Char	I	,	3	E	H	.	*	3	0	o

Table 18 : Index Number Encryption (i Formula) for Test 3

Encrypted Text after Indexing: I,3EH.*30o

Decryption Using Index Formula for Test 3

Encrypted Char	I	,	3	E	H	.	*	3	0	o
Encrypted Index	8	4	3	5	7	1	2	3	0	14
Reverse Formula	$8*2=16$	$(4*2)-3=5$	$(3*2)-3=3$	$(5*2)-3=7$	$(7*2)-3=11$	$(1*2)-3=-1$	$(2*2)/2=1$	$(3*2)-3=3$	$0*2=0$	$(14*2)-3=25$
Original Cipher Char	Q	?	3	H	I	2	@	6	!	Z

Table 19 : Decryption Using Index Formula for Test 3

Decrypted Text after Indexing: Q?3HI2@6!Z

Final Decryption of Test 3

Cipher Text Pair	Decrypted Pair
Q?	Se
3H	cu
I2	re
@6	12

!Z	3x
----	----

Table 20 : Final Decryption of Test 3 Using L-Pattern Playfair Cipher

Final Plain Text: Secure123

4.4 Test 4

Plain Text: Password99

Key: SecureKey!@

Solution: Making an **8×9 matrix** of keyword in **L Pattern** if you read more then click on [Appendix C](#).

Final Plain Text: Password99

4.5 Test 5

Plain Text: Learning AI

Key: CyberSecure123!

Solution: Making an **8×9 matrix** of keyword in **L Pattern**.

8×9 Matrix of Keyword in L Pattern for Test 5

C	y	b	e	r	S	e	c	u
r	e	1	2	3	!	A	B	C
D	F	G	H	I	J	K	L	M
N	O	P	Q	T	U	V	W	X
Y	Z	a	b	d	f	g	h	i
j	k	l	m	n	o	p	q	s
t	v	w	x	y	z	SPACE	.	,
?	+	=	&	%	\$	#	@	*

Table 21 : 8×9 Matrix of Keyword in L Pattern for Test 5

Plain Text Pairs for Test 5

Plain Text	Split into Pairs
Learning AI	Le, ar, ni, ng, <sp> A, I, x*

Table 22 : Plain Text Pairs for Test 5

Cipher Text Table for Test 5

Plain Text Pair	Cipher Text
Le	FQ

ar	bT
ni	Pk
ng	HW
<sp>A	.D
I	I*

Table 23 :Cipher Text Table for Test 5

Final Cipher Text: FQbTPkHW.DI*

Index Number Encryption (i Formula) for Test 5

Rules:

- **Odd index (i):** $(i+3)/2(i+3)/2(i+3)/2$
- **Even index (i):** $i/2i / 2i/2$

Cipher Char	F	Q	b	T	P	k	H	W	.	D	I	*
Original Index (i)	5	16	1	19	15	10	7	22	14	3	8	2
Formula Applied	$(5+3)/2$	$(16/2)$	$(1+3)/2$	$(19+3)/2$	$(15+3)/2$	$(10/2)$	$(7+3)/2$	$(22/2)$	$(14/2)$	$(3+3)/2$	$(8/2)$	$(2/2)$
Encrypted Index	4	8	2	11	9	5	5	11	7	3	4	1
Encrypted Char	D	I	c	K	J	F	F	K	H	D	E	.

Table 24 : Index Number Encryption (i Formula) for Test 5

Encrypted Text after Indexing: DIcKJFFKHD E.

Decryption Using Index Formula for Test 5

Encrypted Char	D	I	c	K	J	F	F	K	H	D	E	.
----------------	---	---	---	---	---	---	---	---	---	---	---	---

Encrypt ed Index	4	8	2	11	9	5	5	11	7	3	4	1
Reverse Formula	(4*2)- 3=5	8*2= 16	(2*2) =4	(11* 2)- 3=19	(9*2)- 3=1 5	(5*2)- 3=7	(5*2)- 3=7	(11* 2)- 3=19	(7*2)- 3=1 1	(3*2)- 3=3	(4*2)- 3=5	1*2 =2
Original Cipher Char	F	Q	b	T	P	k	k	T	H	D	F	b

Table 25 :Decryption Using Index Formula for Test 5

Decrypted Text after Indexing: FQbTPkHW.DI*

Final Decryption of Test 5

Cipher Text Pair	Decrypted Pair
FQ	Le
bT	ar
Pk	ni
HW	ng
.D	<sp>A
I*	I

Table 26 : Final Decryption of Test 5

Final Plain Text: Learning AI

5. Evaluation

This part makes critical assessment on the L Pattern Cryptographic Algorithms, its strengths, weaknesses, and areas of application.

5.1 L Pattern Encryption Algorithm Strengths.

1. Simple and Efficient Design

- The algorithm is simple to comprehend and apply, and thus it is simple to use by beginners and academically.

2. Low Computational Cost

- Low processing power and hence efficient in small systems and low resource environments.

3. Reversible Encryption

- The decryption process can be able to reproduce an original message without losing any data.

4. Pattern-Based Obfuscation

- The basic confidentiality is offered by the L-pattern rearrangement which conceals the original structure of the plain text.

5. Fast Execution Time

- Appropriate in real time applications where encryption and decryption of messages are needed on a fast basis.

5.2 L Pattern Encryption Algorithm Weaknesses.

1. No Protection against Sophisticated Attacks.

- Can be attacked by brute-force and pattern analysis in case of application in high-security settings.

2. No Key-Based Security

- This decreases the overall level of security of the algorithm since it does not have a secret key.

3. Limited Scalability

- Improper with large amounts of data to be secured or with multi-faceted data.

4. Deterministic Encryption Pattern.

- Attackers can identify repetition in encrypted messages as it is fixed.

5. Unsuitable to Military systems or to Banking systems.

- Lacks behind the current cryptography systems such as AES or RSA.

9.3 Uses of L Pattern Encryption Algorithm.

- L Pattern Encryption Algorithm can be successfully implemented in the following aspects:
- Academic and Educational Projects.
- Basic Data Obfuscation
- Cryptography on the Student Level.
- Secure Messaging of Low-Risk Applications.
- Illustration of Encryption Concepts in Training Programmes.

6. Conclusion

The current study was able to design, implement, and test an improved method of encryption using the Playfair cipher and a key matrix in L-pattern and an index-based transformation formula. The paper has shown that classical substitution techniques used with mathematical indexing to enhance the confidentiality and complexity of the generated cipher text is highly effective.

The systematically built algorithm was applied in the development process with initial key matrix generation, rules of pair-based encryption, and, as the final step, the use of the index formula to generate the second security layer. Various test cases were performed with various plaintexts and keys to be sure of correctness, reliability and consistency of both encryption and decryption processes. The findings supported the hypothesis that the suggested technique can extract the original plaintext without the loss of data and generate highly obfuscated cipher text to withstand the easy frequency or pattern analysis.

The analysis of the performance demonstrated efficiency of the algorithm when working with short- and medium-length messages and made it be applicable in practice on educational and low-security applications. The extra transform of the index added more randomness and less predictability than the conventional Playfair cipher. Nevertheless, the technique is still based on symmetric key secrecy and its effectiveness is dependent on safe key management.

Finally, the project fulfilled its goals of studying classical cryptography and improving it with

the new indexing method and testing it in an organized form. It can be improved in the future to make the process automatic via software and incorporate dynamic key generation and test the algorithm against modern attacks on cryptanalytics. This research work offers a solid point of access to the further investigation of hybrid classical-modern encryption methods.

7. References

- Bernstein, C. (2023, April 07). *TechTarget and Informa Tech's Digital Businesses Combine*. Retrieved from <https://www.techtarget.com/searchsecurity/definition/cryptosystem>
- Contributors, W. (2025, Dec 26). Retrieved from Information security: https://en.wikipedia.org/wiki/Information_security?utm_source=chatgpt.com
- GeeksforGeeks. (2025, July 12). *Difference Between Symmetric and Asymmetric Key Encryption*. Retrieved from https://www.geeksforgeeks.org/computer-networks/difference-between-symmetric-and-asymmetric-key-encryption/?utm_source=chatgpt.com
- Hangrae, D. (2024, Sep 18). Retrieved from CSC 316 Cryptography:: https://medium.com/%40dillihangrae/csc-316-cryptography-key-terms-definition-concepts-unveiled-8bf180b7d700?utm_source=chatgpt.com
- Nagaraj, K. (2023, March 11). *Exploring the Playfair Cipher: An Old but Effective Encryption Technique*. Retrieved from Medium: <https://cyberw1ng.medium.com/exploring-the-playfair-cipher-an-old-but-effective-encryption-technique-eadb9696c384>
- Richards, K. (2024, March 13). Retrieved from TechTarget: https://www.techtarget.com/searchsecurity/definition/cryptography?utm_source=chatgpt.com
- Schneider, J. (2025). *A brief history of cryptography*. Retrieved from IBM: https://www.ibm.com/think/topics/cryptography-history?utm_source=chatgpt.com
- Shakil, T. (2014, Nov). *ResearchGate*. Retrieved from An Efficient Modification to Playfair Cipher: https://www.researchgate.net/publication/274709511_An_Efficient_Modification_to_Playfair_Cipher

8. Appendix

Appendix A

Also it is susceptible to known-plaintext attacks, with the fixed rectangular substitution rules occasionally reversible with enough paired plaintext-ciphertext pairs, and typically only 200-300 characters will suffice to hill-climb the cryptanalysis. The fact that the cipher diffuses only to a limited extent, and that changes in a single digit do not propagate to the rest adequately, and that the cipher has no strong confusion properties, are also weaknesses of the cipher, attributable to the predictable 5x5 grid and the lack of any bitwise operations. As an example, during known-plaintext attack, a perpetrator is able to match frequent digraphs to expose the square arrangement of the key, because longer ciphertexts give a chance to differentiate the

patterns. To overcome these, the L-pattern adds L- shaped variations to non-linear patterns and the XOR operations adds diffusion in that a small change of input has a large change of output, similar to the avalanche effect in contemporary ciphers.

Appendix B

After the shift, an XOR operation is performed on derived row and column indices and a cycling sequence based on the ASCII values of the key is used to make sure that the operation modularly wraps around (mod 5) to ensure that the grid remains whole and diffuses changes throughout the ciphertext. This mixed implementation of geometric replacement and bitwise logical operations has better confusion and diffusion characteristics than the original Playfair.

As an illustration, we take a digraph, such as AB, and code (r1=0, c1=3) of A and (r2=1, c2=3) of B in an example key square (even parity of key words). The down shift is first done as they occur in the same column (new r1=1, new r2=2). Then, modified positions by XOR with key values (e.g. 1 and 3) (1 XOR 1=0, 3 XOR 3=0 for one and the other respectively) are looked up to find new letters

Appendix C

8×9 Matrix of Keyword in L Pattern for Test 4

S	e	c	u	r	e	K	e	y
!	@	0	1	2	3	4	5	6
7	8	9	A	B	C	D	F	G
H	I	J	L	M	N	O	P	Q
R	T	V	W	X	Y	Z	a	b
d	f	g	h	i	j	k	l	m
n	o	p	q	s	t	v	w	x
y	z	SPACE	.	,	?	+	=	%

Table 27 :: 8×9 Matrix of Keyword in L Pattern for Test 4

Plain Text Pairs for Test 4

Plain Text	Split into Pairs
------------	------------------

Password99	Pa, ss, wo, rd, 99
------------	--------------------

Cipher Text Table for Test 4

Plain Text Pair	Cipher Text
Pa	S!
ss	@s
wo	3W
rd	Eu
99	5*

*Table 28 : Cipher Text Table for Test 4***Final Cipher Text:** S!@s3WEu5***Index Number Encryption (i Formula) for Test 4****Rules:**

- **Odd index (i):** $(i+3)/2$ $(i+3)/2$ $(i+3)/2$
- **Even index (i):** $i/2$ $i/2$ $i/2$

Cipher Char	S	!	@	s	3	W	E	u	5	*
Original Index (i)	18	0	1	18	3	22	4	20	5	6
Formula Applied	$(18/2)$	$(0/2)$	$(1+3)/2$	$(18/2)$	$(3+3)/2$	$(22/2)$	$(4/2)$	$(20/2)$	$(5+3)/2$	$(6/2)$
Encrypted Index	9	0	2	9	3	11	2	10	4	3
Encrypted Char	J	SPACE	*	J	3	M	*	K	E	D

*Table 29 : Index Number Encryption (i Formula) for Test 4***Encrypted Text after Indexing:** J *JJ3M*KED**Decryption Using Index Formula for Test 4**

Encrypted Char	J	SPACE	*	J	3	M	*	K	E	D
Encrypted	9	0	2	9	3	11	2	10	2	3

Index										
Reverse	(9*2)	0*2=0	(2*2)=	(9*2)	(3*2)	(11*2)	2*2=	(10*2)	2*2=	(3*2)
Formula	-		4	-	-3=3	-3=19	4	-3=17	4	-3=3
	3=15			3=15						
Original	S	!	@	s	3	W	*	u	E	D
Cipher										
Char										

Table 30 : Decryption Using Index Formula for Test 4

Decrypted Text after Indexing: S!@s3WEu5*

Final Decryption of Test 4

Cipher Text Pair	Decrypted Pair
S!	Pa
@s	ss
3W	wo
Eu	rd
5*	99

Table 31 : Final Decryption of Test 4